

Etching and deposition

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1 Etching

Etching refers to the removal of material from the wafer surface. The process is usually combined with lithography in order to select specific areas on the wafer from which material is to be removed. Etching represents one way of permanently transferring the mask pattern from the photoresist to the wafer surface. The complementary process to etching is *deposition* (or growth), where new material is added. Unlike oxidation (or nitridation), where the underlying Si is consumed to form the oxide (nitride) layer, in deposition, new material is added without consuming the underlying wafer.

There are two main types of etching

1. Wet etching

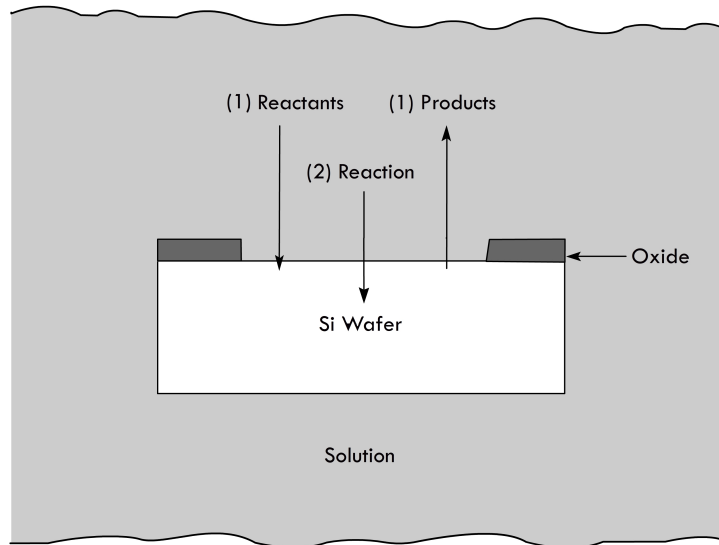


Figure 1: Schematic of the wet etching process. A controlled portion of the wafer surface is exposed to the etchant which then removes materials by chemical reaction. Adapted from *Fundamentals of semiconductor manufacturing and process control* - May and Spanos.

2. Dry etching

1.1 Wet etching

In wet etching, the wafers are immersed in a tank of the etchant (mix of chemicals), as shown in figure 1. There is a chemical reaction between the wafer surface and the etchants that helps in material removal. Either a photoresist layer or a hard mask like oxide or nitride layer is used to protect the rest of the wafer. The time for etching depends on the amount and type of material that needs to be removed. KOH (potassium hydroxide) is a common etchant used to remove Si. Usually, 30% KOH solution is used, which has a etch rate of $\sim 100 \mu\text{m/hr}$ at 90°C . Thus, an entire 4" wafer, with thickness of $500 \mu\text{m}$, can be etched through in approximately 5 hours. The etch rate of Si (100) by 30 % KOH is shown in figure 2. After etching, the wafers are rinsed, usually in DI water, for removal of etchant and then finally dried.

Wet etching is used for removal of material from large areas (trench sizes $> 3 \mu\text{m}$). For smaller areas, where greater precision in removal of material is required, dry etch is preferred. The wet etching process is anisotropic i.e. the etch rate depends on the plane of the Si wafer, from which atoms are being

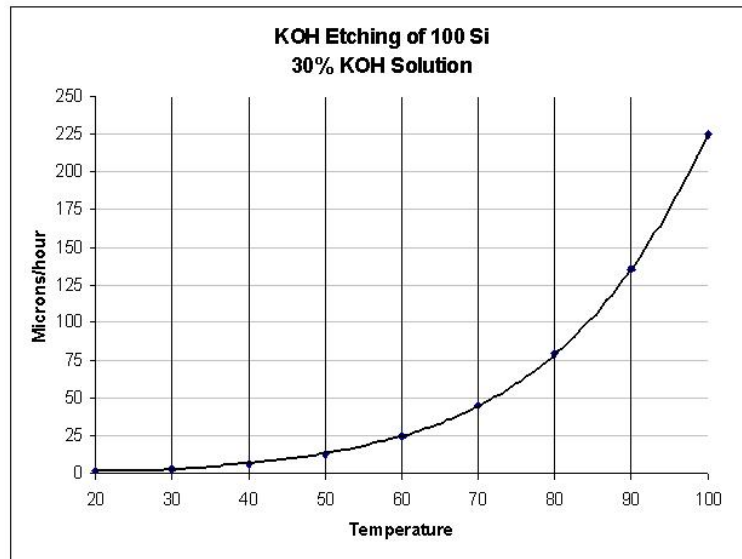


Figure 2: Etch rate of Si in KOH as a function of temperature. There is a non-linear increase in etch rate with increasing temperature. Typically, etching is carried out at 90 °C. Source <http://www.cleanroom.byu.edu/KOH.phtml>

removed. The etch rate for Si (110), in the same 30 % KOH, is shown in figure 3. Compared to Si(100) plane, figure 2, the rate is higher. This means that wet etching of Si(100) will produce a trapezoidal profile, with a specific angle of 54.74 deg, as shown in figure 4. Etching uniformity is important to get a uniform thickness over the entire wafer surface. This is usually determined by process conditions like etchant temperature, concentration, and agitation (using stirrers).

1.2 Etching challenges

There are some process challenges related to etching. These are common to both wet and dry etching, though they are more pronounced and harder to control in wet etching due to the higher rate of material removal, compared to dry etching.

1.2.1 Incomplete etch

In incomplete etch, the time is not sufficient for complete material removal. This is usually due to concentration or temperature not being sufficient. The concentration profile left behind is usually a rough surface, due to local variations in material removal, as depicted in figure 5.

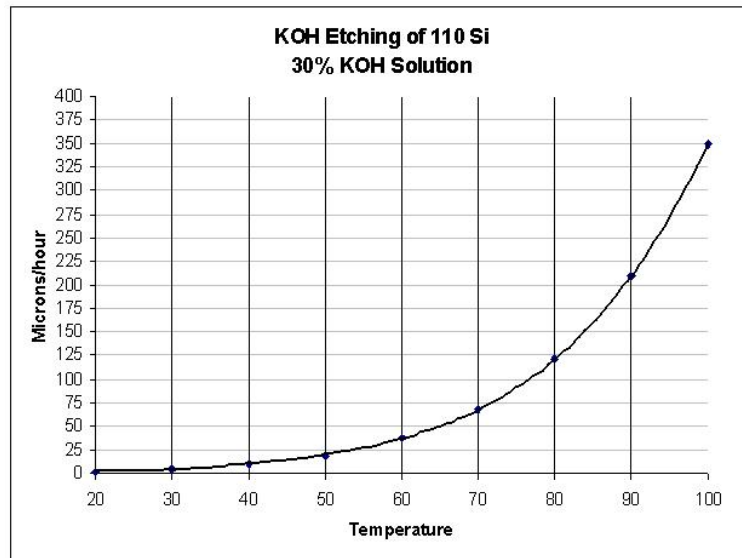


Figure 3: Etch rate of Si(110) in KOH as a function of temperature. Compared to the etch rate for Si (100), see figure 2, the etch rate is higher. Source <http://www.cleanroom.byu.edu/KOH.phtml>

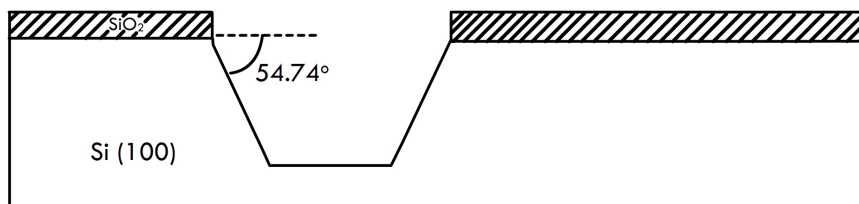


Figure 4: Anisotropic etching of Si by KOH. Because of the difference in the etch rates of Si along the different crystallographic layers the final profile is trapezoidal, with the angle determined by the etch rates. Source <http://www.cleanroom.byu.edu/KOH.phtml>

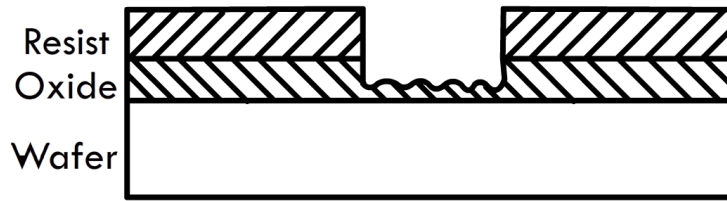


Figure 5: Surface profile for incomplete etch of oxide layer on Si. A resist layer protects the remaining oxide. A rough oxide layer is left behind due to the local variations in rate of removal of the oxide layer. Adapted from *Microchip fabrication* - Peter van Zant.

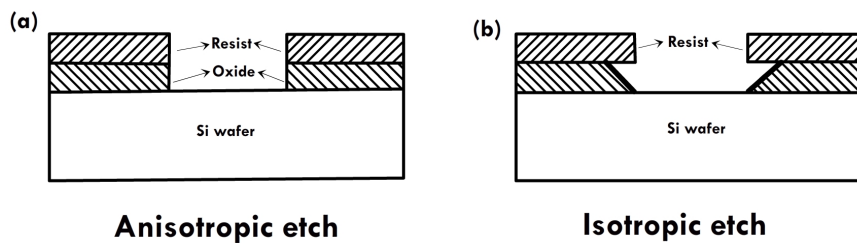


Figure 6: (a) A complete anisotropic etching produces vertical side walls. (b) Most often etching is partially isotropic, so that side walls are formed at an angle. Adapted from *Microchip fabrication* - Peter van Zant.

1.2.2 Over etch and undercutting

The opposite of incomplete etching is over etching. An ideal etchant is selective and completely anisotropic. This is essential to get vertical sidewalls when a trench is created. But, this is not always possible, so that sloped side walls are obtained, see figure 6. When the etch time is larger than the required etch time, due to isotropic etching, material under the photoresist can get removed. This is called *over etching* and in extreme cases it can also lead to liftoff of the resist layer, see figure 7. This is harmful, since it exposes areas of the wafer that the resist protects to the etching process.

1.2.3 Etch selectivity

Etching process should be selective to the material that has to be removed. This helps to protect the material under the mask (within limits of isotropic etching) and also the mask material itself (oxide, nitride, or resists). Consider

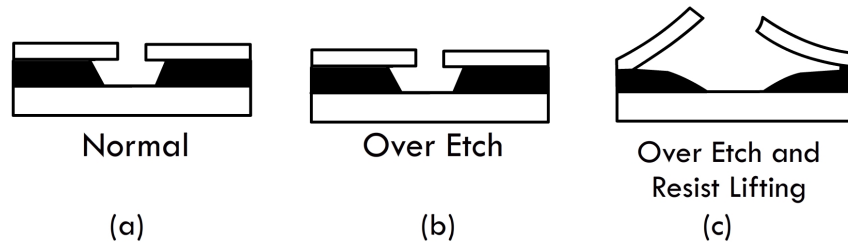


Figure 7: (a) Normal (b) Over etching and (c) Resist liftoff due to excess over etching. The etching rate and time are crucial to prevent over etching since resist removal can cause damage to portions of the wafer that have to be protected from the etchant. Adapted from *Microchip fabrication - Peter van Zant*.

Si etching, using KOH. The etch rate for Si(100) at 90 °C using 30 % KOH is $\sim 100 \mu\text{m/hr}$, see figure 2. If silicon nitride is used as mask, its etch rate, under the same conditions, is $\sim 1 \text{ nm/hr}$, nearly 10^5 times slower than Si. Thus, silicon nitride is commonly used as a mask for Si etching (especially for making Si cantilever based devices). On the other hand, silicon dioxide etch rate, under the same conditions, is $\sim 1 \mu\text{m/hr}$, see figure 8. So, using silicon oxide as a mask will not be good enough or a very thick oxide layer is required.

Different etchants that are used for different layers and the corresponding etch rates, shown in table1. For Si etching, KOH is used or a mixture of nitric acid and hydrofluoric acid (HF). For silicon oxide etching, usually a mixture of HF and ammonium fluoride (NH_4F) is used, that produces a etch rate of $\sim 0.1 \mu\text{m/hr}$ at room temperature. This mixture does not etch Si, so it provides very good selectivity. This etchant is called *BOE* (buffered oxide etchants). For silicon nitride, usually a strong acid like hot phosphoric acid is used at high temperatures (180 °C) since it is a very good passivating layer and hard to remove under normal conditions.

2 Dry etching

Dry etching, as the name suggest, is removal of material in the absence of solvent. The process was introduced because wet etching has some limitations in its applicability, which are listed below.

1. Wet etching is used for large pattern sizes, usually larger than $2 \mu\text{m}$.
2. It is an isotropic process - sloped sidewalls rather than straight walls.

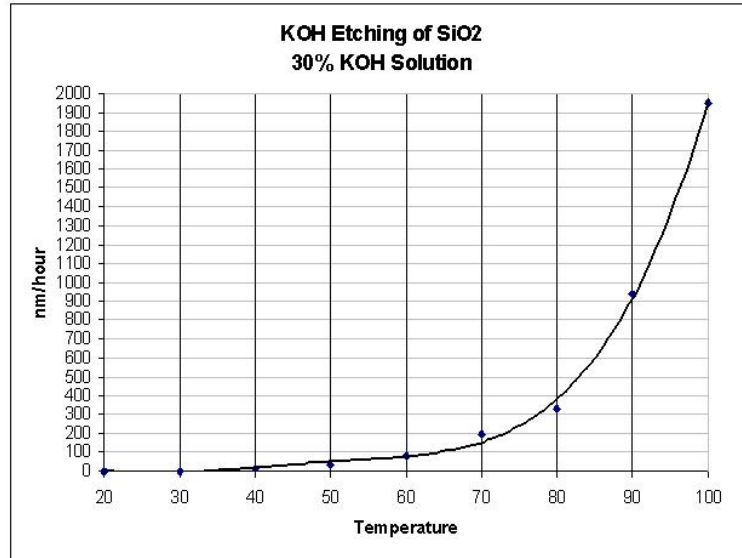


Figure 8: Etch rate of SiO₂ by KOH. Because of comparable etch rates to pure Si, it cannot be used as an etch mask for Si etching. Silicon nitride is used as the mask since its etch rate, under the same conditions, is of the order of nm/hr. Source <http://www.cleanroom.byu.edu/KOH.phtml>

Table 1: Etching chemicals used for different layers and their etch rates, under commonly used conditions. Adapted from *Microchip fabrication - Peter van Zant*.

Material	Common etchant	Etch temperature	Etch rate (Å/min)
SiO ₂	HF NH ₄ F (1:8)	Room temperature	700
SiO ₂	Acetic acid NH ₄ F (2:1)	Room temperature	1000
Aluminum	HPO ₄ HNO (nitroxyl) Acetic acid water	40-50 °C	2000
Si ₃ N ₄	H ₃ PO ₄	150-180 °C	80
Poly Si	HNO ₃ H ₂ O HF	Room temperature	1000

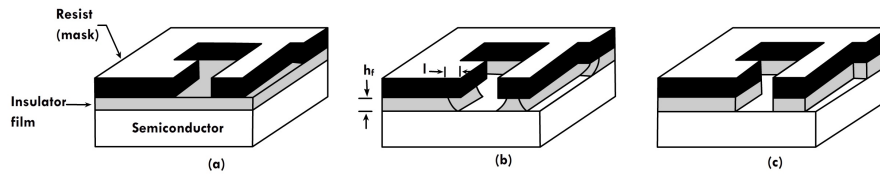


Figure 9: (a) Starting surface after development of the resist (b) Surface after wet etching (c) Surface after dry etching. Because of the anisotropic nature of etching, dry etching produces more vertical side walls compared to wet etching, but the removal rate is slower. Adapted from *Fundamentals of semiconductor manufacturing and process control* - May and Spanos.

3. Wet etch has to be combined with subsequent rinse and dry steps. This increases chances of defects or contamination.
4. Hazardous chemicals and conditions are used, so safety is an issue. Safe disposal of chemicals is essential.
5. Undercutting and resist peel off can happen if time is not controlled or etch conditions change during process.

The wet and dry etching process are compared in figure 9. Dry etching is a process that overcomes some of these issues. Here, *etchant gases* are the primary medium for the removal of material. The basic steps involved are summarized in figure 10. There are three main types of dry etching

1. Plasma etch
2. Ion beam milling
3. Reactive ion etch

2.1 Plasma etch

In plasma etch, the chemical etchant is introduced in the gas phase. For etching silicon oxide, CF_4 (tetrafluoromethane) is used. The chamber is first evacuated before introducing the gas. Radio frequency (RF) electrodes are then used to generate the plasma that ionizes the gas. This ionized gas attacks the oxide layer, removing the layer. Etch rates in plasma etch are $\sim 1 - 10 \mu\text{m/hr}$, much smaller than wet etching. So, it more suitable for thin layers, but it also provides greater thickness control. There are different configurations for plasma etching, one such *planar configuration* is shown in

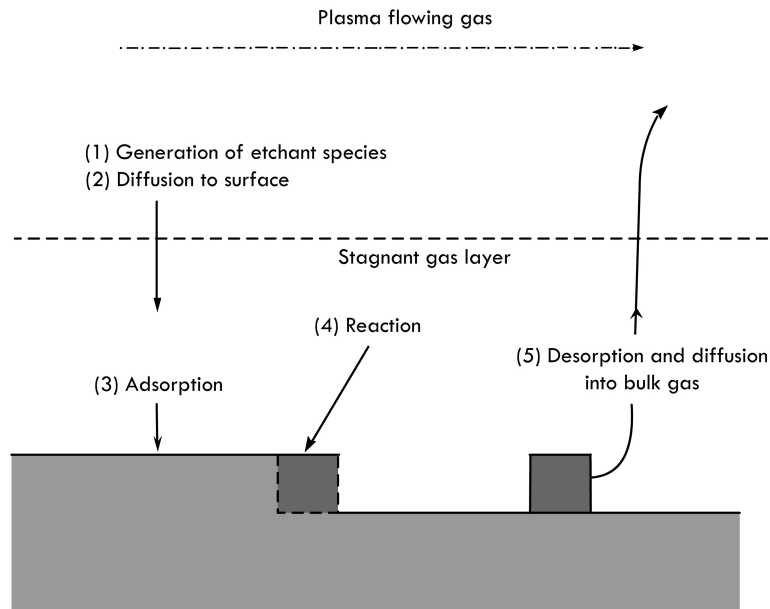


Figure 10: Various steps from (1) - (5) in the dry etch process. Gases are transported to the wafer surface, where they adsorb and react with the wafer surface material, at the step edges. The gases then desorb from the surface. Adapted from *Fundamentals of semiconductor manufacturing and process control* - May and Spanos.

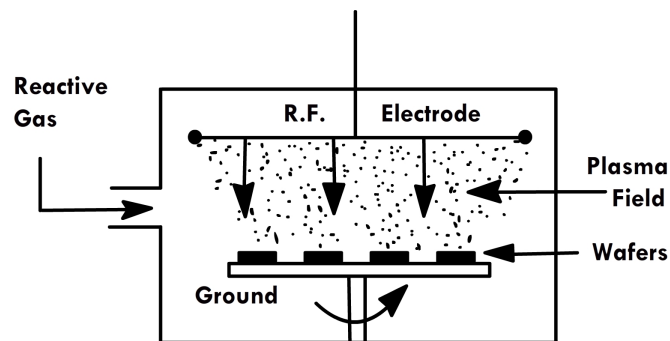


Figure 11: Planar plasma etch configuration. The wafers are held on a grounded chuck, close to the RF electrodes. Reactive gas introduced in the chamber, is ionized and the ions help in material removal. Adapted from *Microchip fabrication* - Peter van Zant.

Table 2: Typical plasma etching chemicals for different film materials and the corresponding gaseous products. Adapted from *Microchip fabrication* - Peter van Zant.

Film	Etchant	Typical gas compounds
Al	Chlorine	BCl_3 , CCl_4 , Cl_2 , SiCl_4
Mo	Fluorine	CF_4 , SF_4 , SF_6
Polymers	Oxygen	DF_4 , SF_4 , SF_6
Si	Chlorine	BCl_3 , CCl_4 , Cl_2 , SiCl_4
	Fluorine	CF_4 , SF_4 , SF_6
SiO_2	Fluorine	CF_4 , CHF_3 , C_2F_6 , C_3F_8
Ta	Fluorine	CF_4 , CHF_3 , C_2F_6 , C_3F_8
Ti	Fluorine	CF_4 , CHF_3 , C_2F_6 , C_3F_8
W	Fluorine	CF_4 , CHF_3 , C_2F_6 , C_3F_8

figure 11. The resist layer used to protect the wafer is also etched along with the oxide. But the resist thickness is much larger than the oxide (few μm of resist compared to tens of nm of oxide). This means that substantial amount of resist is still available, after the etching process. Some of the different etchant gases used for plasma etching of various films are shown in table 2.

2.2 Ion beam etch

Ion beam etching is similar to the ion beam milling process that is used for transmission electron microscopy sample preparation. This is a physical process where ionized inert gas ions (usually Ar) are used to remove material from the wafer. The process is *not selective* but it is *highly directional*. The ion beam etching process is shown in figure 12.

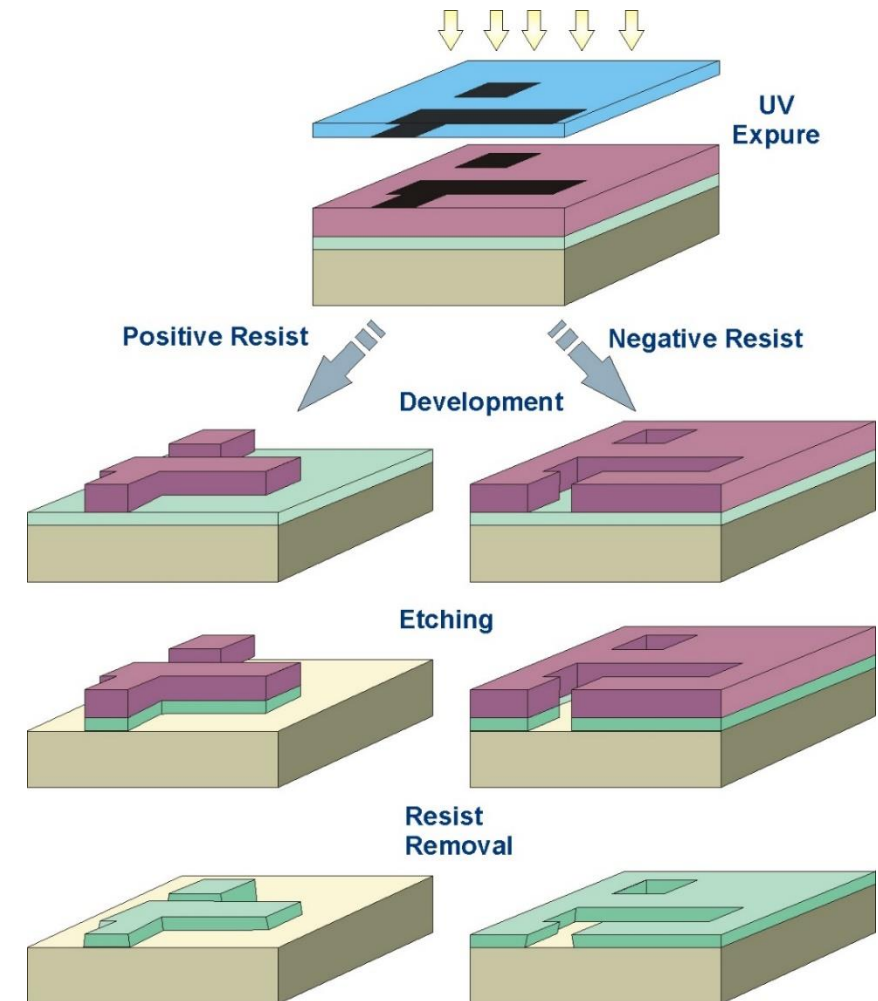
2.3 Reactive ion etching

Reactive ion etching combines the plasma and ion beam etching process to achieve both selectivity and directionality. There is an increase in selectivity compared to plasma etch, for SiO_2 and Si the selectivity ratio is 35:1 while for pure plasma etch the ratio is 10:1. This reduces the thickness requirement on the mask. Dry etch process is also used for resist stripping after patterning is complete. This is usually done by plasma etching using oxygen.

Etching is a process by which material is removed from a substrate or from thin films on the substrate surface. When a mask layer is used to protect specific regions of the wafer surface, the goal of the etching is to «precisely» remove the material which is not covered by the mask

There are two fundamental groups of etching:

- **wet etching** (liquid-based etchants)
- **dry etching** (plasma-based etchants, but not only ...)



In general an ideal etch process is not completely attainable

Etching processes are not capable of precisely transferring the pattern established by a mask.

PARAMETERS

BIAS. Difference in lateral dimension between the etched image and the mask

ETCH RATE. The rate at which the material is removed.

SELECTIVITY. ratio of the etch rates of one material over the other. For example, the selectivity is ratio of the etch rate of the layer being etched to the etch rate of the mask or the layer under the layer being etched. Etching with high selectivity is supposed to remove the selected layer entirely without harming the substrate and mask.

ANISOTROPY

- Isotropic etching has the same etch rate in all directions.
- Anisotropic etching has different etch rates in the lateral and vertical directions.

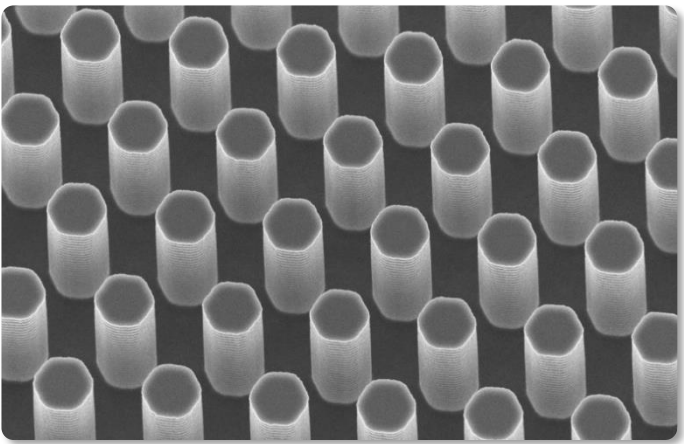
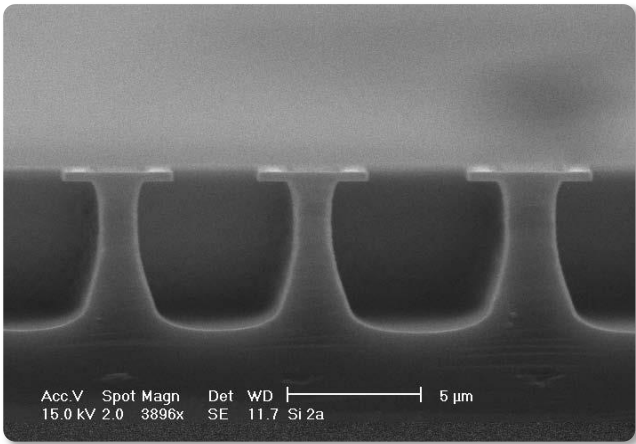
Anisotropic etching is preferable in semiconductor manufacturing processes and for **micro- nano fabrication**. Most wet etching profiles are isotropic, except for etching crystalline materials, whereas etching profiles of dry etching are anisotropic.

Anisotropy is defined as

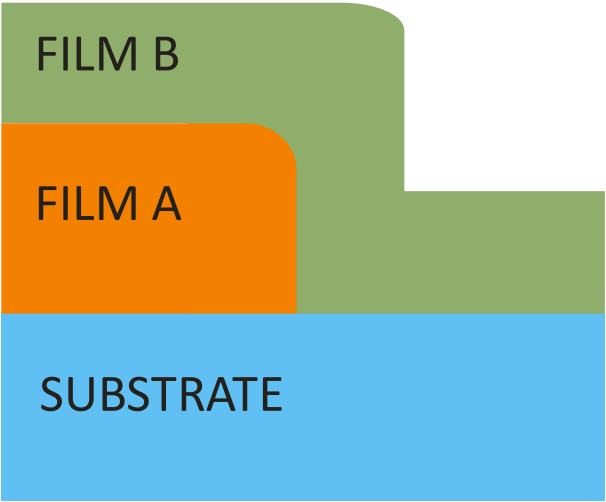
$$A = 1 - \frac{R_L}{R_V},$$

where R_L and R_V are the lateral and vertical etch rates, respectively. For perfectly anisotropic etching, A is 1. For perfectly isotropic etching, A is 0.

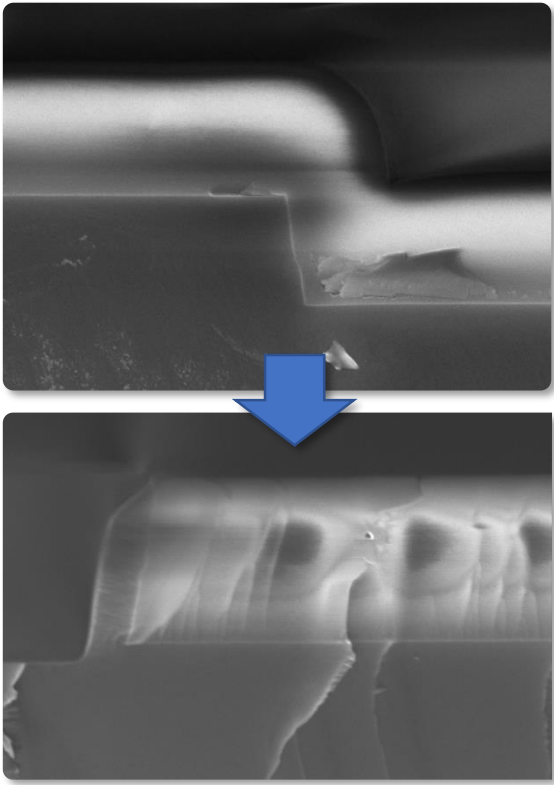
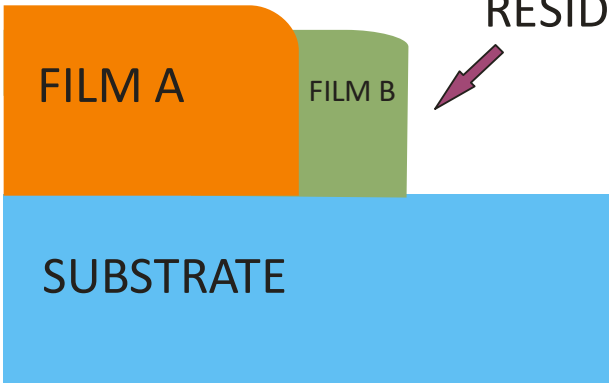
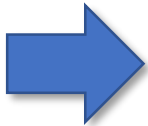
AN EXAMPLE



ISOTROPIC (LEFT) AND ANISTROPIC (RIGHT)

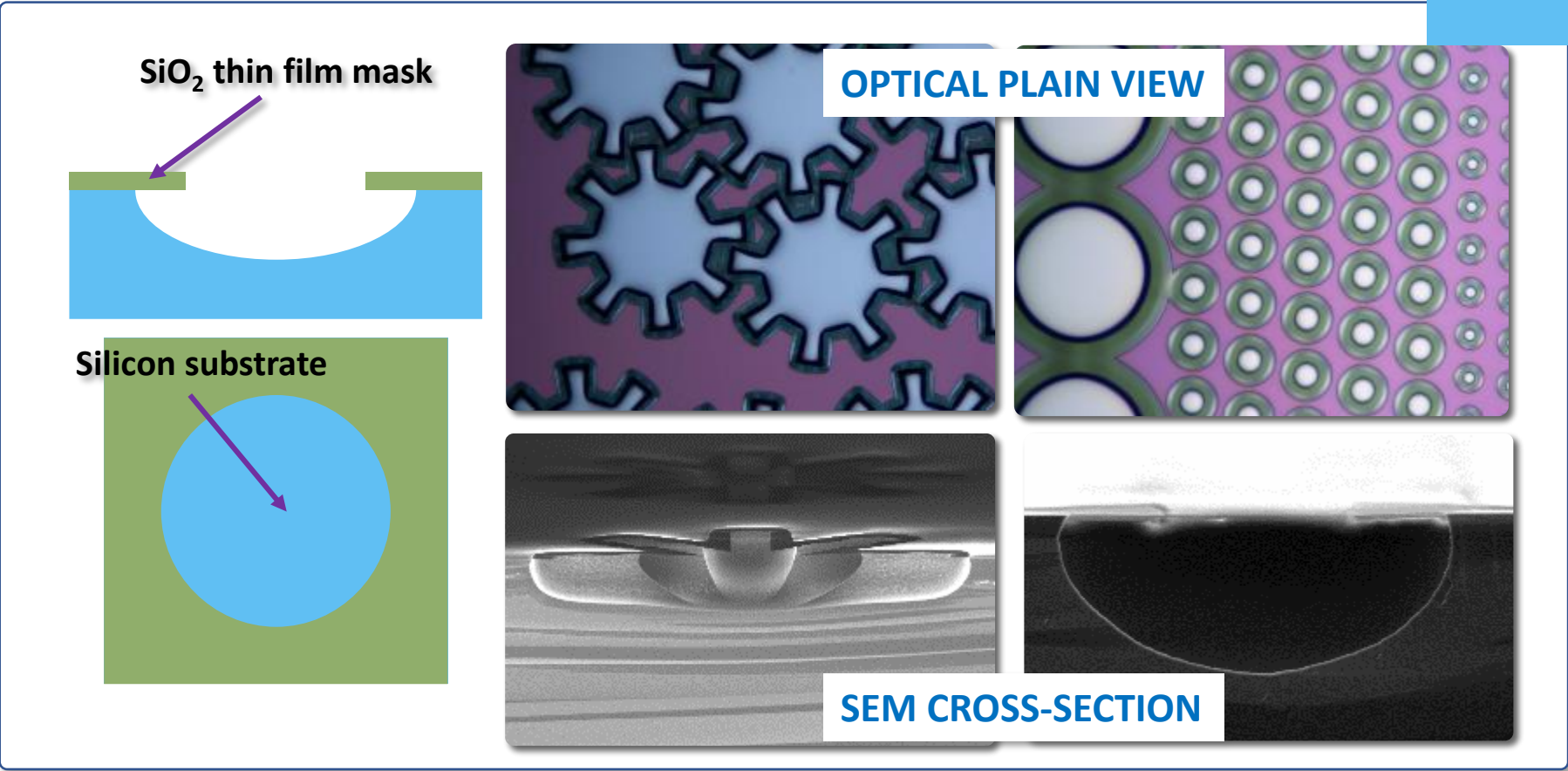
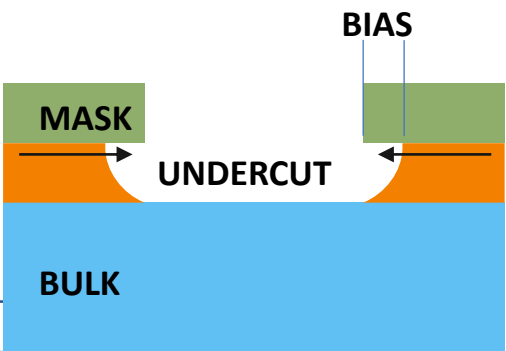


ANISOTROPIC
ETCHING



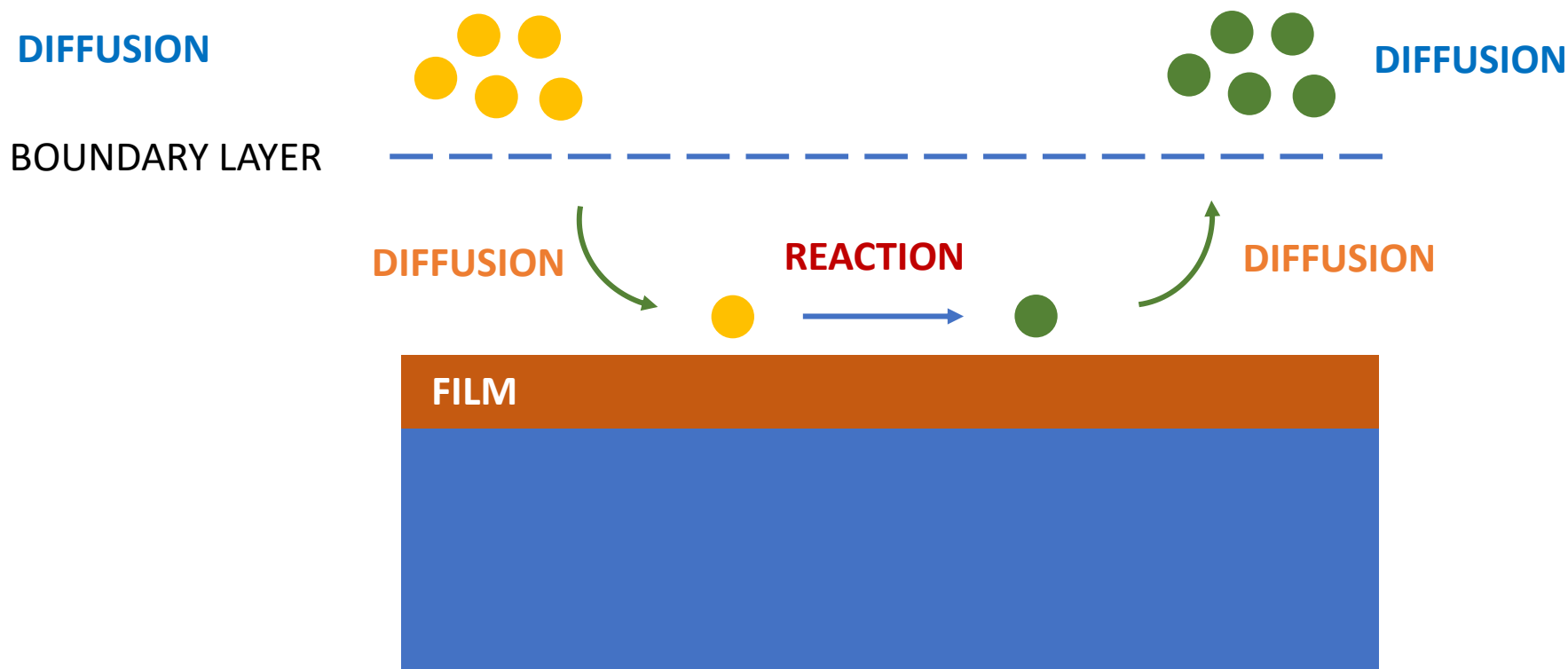
If etching is anisotropic, overetching is needed to remove residual materials at step

UNDERCUT. Lateral distance per side under the mask, as shown in Figure 1. It can be characterized by the etch rate anisotropy A. Sometimes, the etchant will attack the photoresist pattern, which causes an etch bias.



ETCHING PROCESS CONSIST OF THREE STEPS

- Mass transport of reactants (through a boundary layer) to the surface to be etched
- Reaction between reactants and film surface to be etched at the surface
- Mass transport of the reaction products from the surface through the surface boundary layer



- Wet etching was used exclusively till 1970's when **feature size >3um**.
- For small scale features, **large etch bias** leads to significant CD (critical dimension) loss.
- For today's IC industry, wet etching is used for noncritical feature sizes.

ADVANTAGES

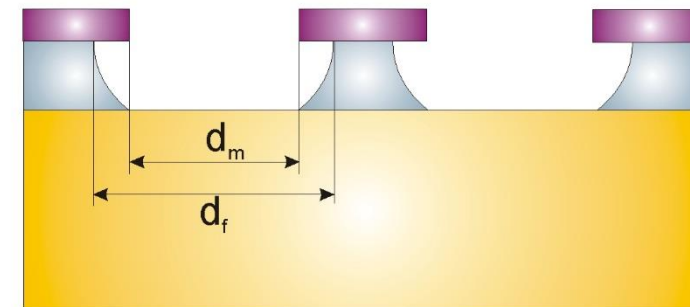
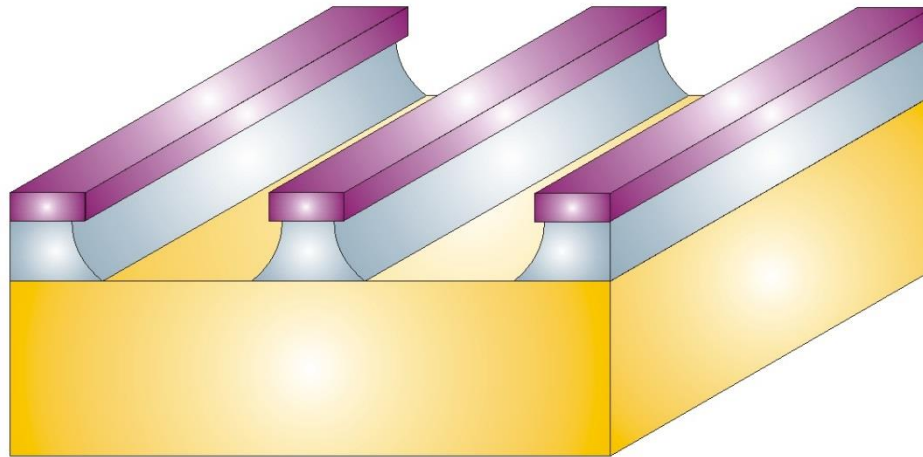
high selectivity, relatively inexpensive equipment, batch system with high throughput, etch rate can be very fast (many $\mu\text{m}/\text{min}$).

DISADVANTAGES

generally isotropic profile, high chemical usage, poor process control (not so reproducible), excessive particulate contamination.

- The etch rate can be controlled by any of the three serial processes: **reactants transport** to the surface (depends on chemical concentration and stirring...), **reaction rate** (depends on temperature), **reaction products transport** from the surface (depends on stirring...).
- Preference is to have **reaction rate** controlled process because
 - Etch rate can be increased by temperature
 - Good control over reaction rate – temperature of a liquid is easy to control
- Mass transport control will result in non-uniform etch rate: edge etches faster.
- Etchant is often stirred to minimize boundary layer and make etching more uniform.

SiO₂	HF, HF + NH ₄ F
Si₃N₄	BOILING 85% H ₃ PO ₄ @ 180 °C
POLY/SILICON	H ₂ O+ HNO ₃ + HF
ALUMINUM	H ₃ PO ₄ +HNO ₃ +H ₂ O @ 30 °C
GOLD	KI + I ₂ + H ₂ O
TITANIUM	HF DILUTED
PLATINUM	H ₂ O + HCl + HNO ₃ (ACQUA REGIA)



Wet and Dry Etching

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Isotropy and anisotropy:

When a material is attacked by a liquid or vapor etchant, it is removed isotropically (uniformly in all directions) or anisotropic etching (uniformity in vertical direction). The difference between isotropic etching and anisotropic etching is shown in Figure 1. Material removal rate for wet-etching is usually faster than the rates for many dry etching processes and can easily be changed by varying temperature or the concentration of active species.

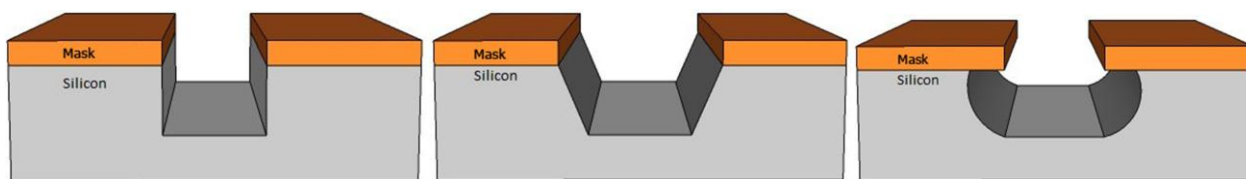


Figure 1. (a) Completely anisotropic (b) Partially anisotropic and (c) Isotropic etching of silicon

Wet Etch

Synonyms: chemical etching, liquid etching

Definition:

Wet etching is a material removal process that uses liquid chemicals or etchants to remove materials from a wafer. The specific patterns are defined by masks on the wafer. Materials that are not protected by the masks are etched away by liquid chemicals. These masks are deposited and patterned on the wafers in a prior fabrication step using lithography.^[2]

A wet etching process involves multiple chemical reactions that consume the original reactants and produce new reactants. The wet etch process can be described by three basic steps. (1) Diffusion of the liquid etchant to the structure that is to be removed. (2) The reaction between the liquid etchant and the material being etched away. A reduction-oxidation (redox) reaction usually occurs. This reaction entails the oxidation of the material then dissolving the oxidized material. (3) Diffusion of the byproducts in the reaction from the reacted surface.

Anisotropic wet etching:

Liquid etchants etch crystalline materials at different rates depending upon which crystal face is exposed to the etchant. There is a large difference in the etch rate depending on the silicon

crystalline plane. In materials such as silicon, this effect can allow for very high anisotropy. Some of the anisotropic wet etching agents for silicon are potassium hydroxide (KOH), ethylenediamine pyrocatechol (EDP), or tetramethylammonium hydroxide (TMAH). Etching a (100) silicon wafer would result in a pyramid shaped etch pit as shown in Figure 2a. The etched wall will be flat and angled. The angle to the surface of the wafer is 54.7° . Figure 2c-d depicts scanning electron micrographs of (110)-oriented two-dimensional silicon walls with micro and nanoscale dimensions generated based on KOH based wet etching.

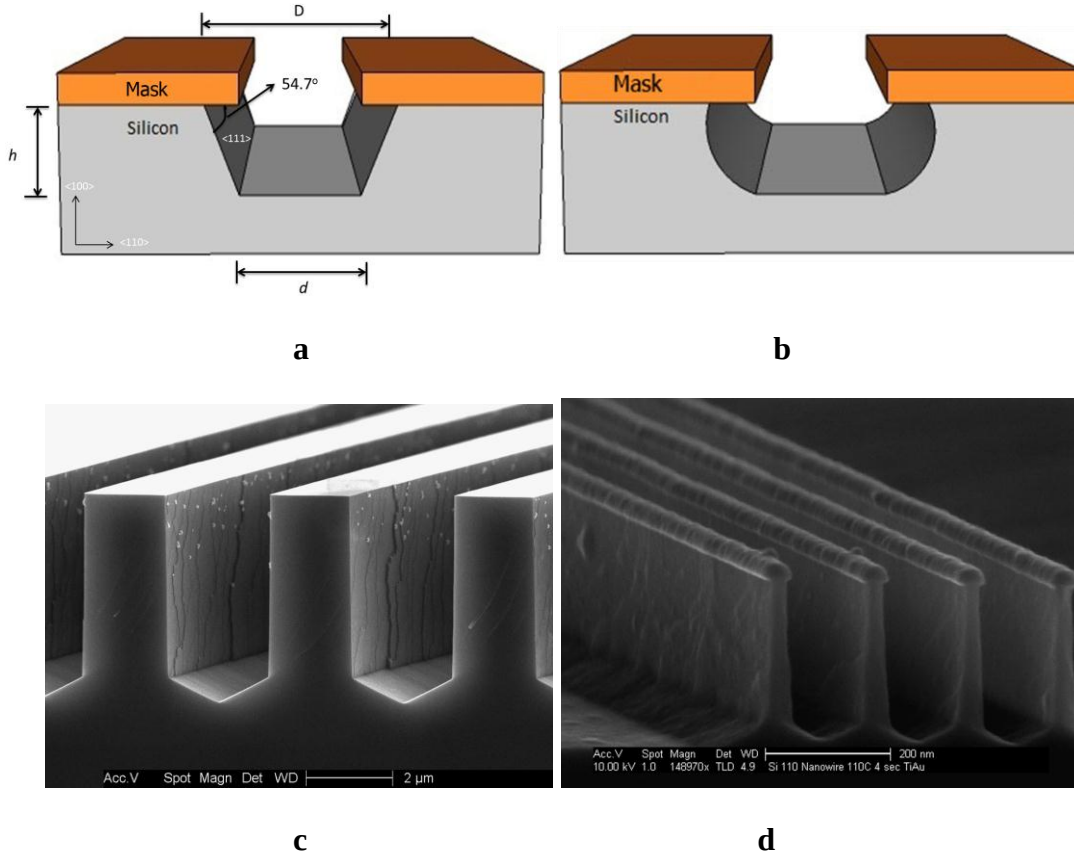


Figure 2. Schematics of an etch profile in (a) an anisotropic and (b) an isotropic etch of a (100) oriented silicon surface. (c-d) KOH based wet etching of (110)-oriented Si surfaces with micro and nanoscale two-dimensional walls.

The relationship between mask dimensions, etch depth and the floor width is given in equation 1.

$$d = D - \left(\frac{2h}{\tan(54.7^\circ)} \right) \text{ [equation 1]}$$

Isotropic wet etching:

For isotropic wet etching, a mixture of hydrofluoric acid, nitric acid, and acetic acid (HNA) is the most common etchant solvent for silicon. The concentrations of each etchant determines the etch rate. Silicon dioxide or silicon nitride is usually used as a masking material against HNA. As the reaction takes place, the material is removed laterally at a rate similar to the speed of etching downward. This lateral and downward etching process takes place even with isotropic dry etching which is described in the dry etch section.

Wet chemical etching is generally isotropic even though a mask is present since the liquid etchant can penetrate underneath the mask (Figure 2b). If directionality is very important for high-resolution pattern transfer, wet chemical etching is normally not used.

Dry Etch:

Synonyms: plasma etching, gas etching, physical dry etching, chemical dry etching, physical-chemical etching

Definition:

In dry etching, plasmas or etchant gasses remove the substrate material. The reaction that takes place can be done utilizing high kinetic energy of particle beams, chemical reaction or a combination of both.

Physical dry etching:

Physical dry etching requires high energy kinetic energy (ion, electron, or photon) beams to etch off the substrate atoms. When the high energy particles knock out the atoms from the substrate surface, the material evaporates after leaving the substrate. There is no chemical reaction taking place and therefore only the material that is unmasked will be removed. The physical reaction taking place is illustrated in Figure 3.

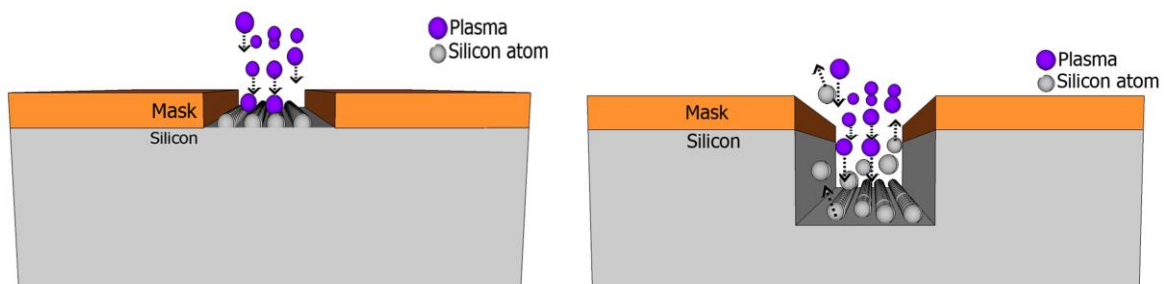


Figure 3. The plasma hits the silicon wafer with high energy to knock-off the Si atoms on the surface. (a) The plasma atoms hitting the surface. (b) The silicon atoms being evaporated off from the surface.

Chemical dry etching:

Chemical dry etching (also called vapor phase etching) does not use liquid chemicals or etchants. This process involves a chemical reaction between etchant gases to attack the silicon surface. The chemical dry etching process is usually isotropic and exhibits high selectivity. Anisotropic dry etching has the ability to etch with finer resolution and higher aspect ratio than isotropic etching. Due to the directional nature of dry etching, undercutting can be avoided. Figure 4 shows a rendition of the reaction that takes place in chemical dry etching. Some of the ions that are used in chemical dry etching is tetrafluoromethane (CF_4), sulfur hexafluoride (SF_6), nitrogen trifluoride (NF_3), chlorine gas (Cl_2), or fluorine (F_2).^[3]

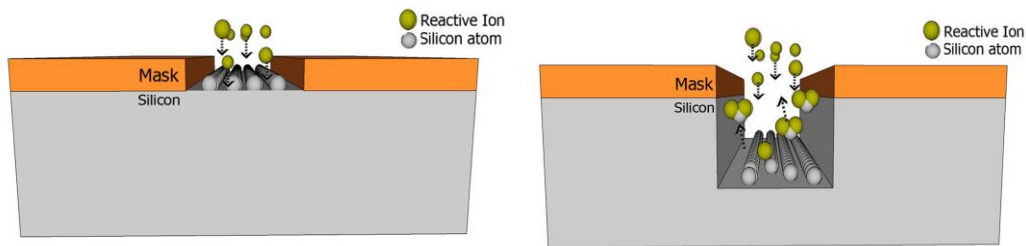


Figure 4. Process of a reactive ion interacting with the silicon surface. (a) The interaction between the reactive ion and the silicon atom. (b) A bond between the reactive ion and the silicon atom then chemically remove the silicon atoms from the surface.

Reactive Ion Etching:

Reactive ion etching (RIE) uses both physical and chemical mechanisms to achieve high levels of resolution. The process is one of the most diverse and most widely used processes in industry and research. Since the process combines both physical and chemical interactions, the process is much faster. The high energy collision from the ionization helps to dissociate the etchant molecules into more reactive species.

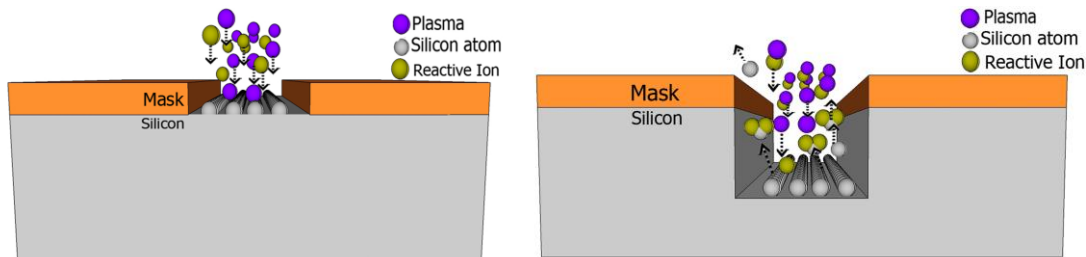


Figure 5. The RIE process. This process involves both physical and chemical reactions to etch off the silicon.

In the RIE-process, cations are produced from reactive gases which are accelerated with high energy to the substrate and chemically react with the silicon. The typical RIE gasses for Si are CF_4 , SF_6 and $\text{BCl}_2 + \text{Cl}_2$. As seen in Figure 5, both physical and chemical reaction is taking

place. Figure 6 depicts some micro/nano structures with high aspect ratio etched using RIE process.

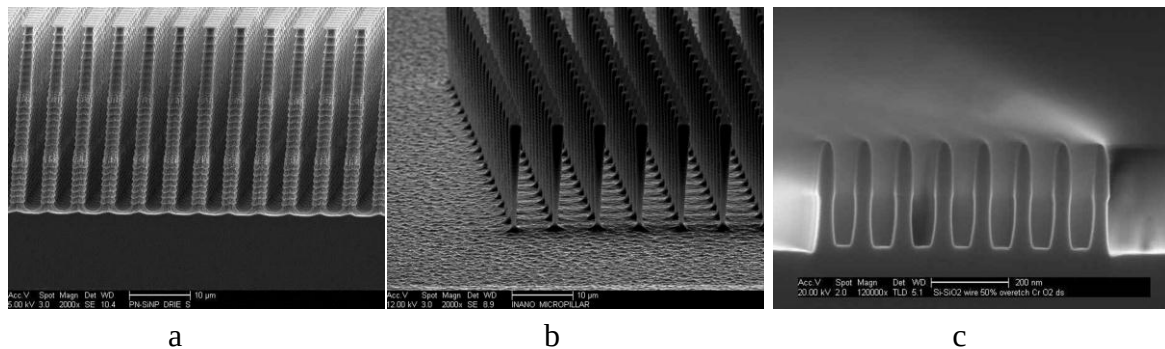


Figure 6. (a-b) Silicon micro-pillars fabricated using deep reactive ion etching (DRIE). These pillars were made using a Bosch process. The Bosch process is a pulsed-multiplexed etching technique which alternates between two modes to achieve extremely long and vertical micron-scaled nanowires.^[1] (c) SiO₂ and Si nanowalls etched via RIE process.

An extensive study was done to determine the dependence of the etching performance by Chen et al.^[4] Factors such as applied coil or electrode power, reactant gas flow rates, duty cycles, and chamber pressures were considered. These process parameters were modeled and experimentally determined to see the effect on the surface morphology and the mechanical performance of the silicon structures.

References:

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